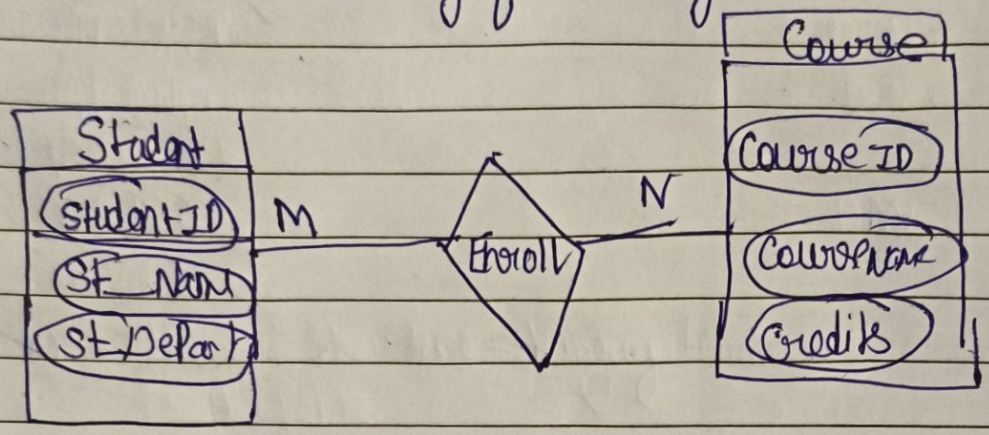
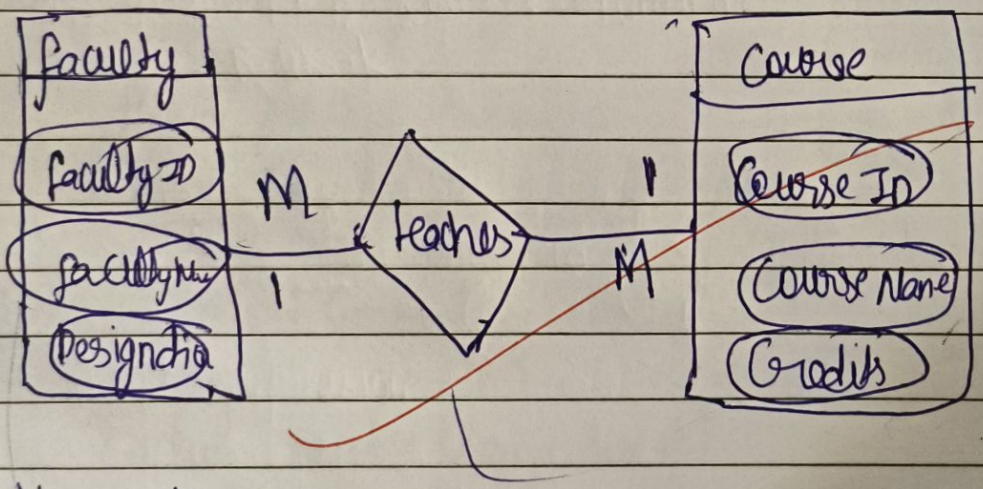
DBMS

Yashsharma Assignment-1 2401331540207

Answer:

ER modeling for university database





b) mapping constraints
- Student - Course : Many to many
  Course - faculty - Many to one
  faculty - course one to many

c) Relational tables
- Student (Student ID, Name, Department)
- Faculty (Faculty ID, Faculty Name, Designation)
- Course (Course ID, Course Name, Credits)

Answers2

a) Required type
Logical Data Independence

b) Justification

Adding a new attributes (email) changes the logical schema but should not affect application programs

c) Example of Physical Data Independence

changing storage from hard disk to SSD without affecting database queries

Answer 3

c) Operators used

i) Selection (σ)
ii) Projection (π)
iii) Join ($\bowtie$)
iv) Set Difference ($-$)

Answer 4.

Statement	Category
CREATE TABLE	DDL
INSERT	DML
GRANT	DCL
ROLLBACK	TCL
UPDATE	DML

b) Database Structure-changing Command
Create table

c) Transaction controll Command
Rollback

Answer 5

Indexing and Query Optimization

a) Cost Comparision

- SELECT * from Employee WHERE DeptNO=10
→ use Index faster access, low cost

- SELECT * FROM EMPLOYEE
→ full table Scan high cost

b) optimized Query

select * from employee where DeptNo = 10

c) Role of Indexing

Indexing reduces risk I/O operations, speeds up and improve query Execution time.

Answer 6

Give FDs

$A \to B$
$B \to C$
$A \to D$
$D \to E$

a) ~~Closure~~ attribute Closure

$A^+ = \{A, B, C, D, E\}$

b) Candidate Key

A

c) Superkeys - A, AB, AC, AE, ABC etc

Date
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Answer 7

Normalization

Given relation

Student_course (RollNo, StudentName, CourseID, CourseName, faculty)

a) Dependencies
- Partial Dependency : RollNo → StudentName
- Transitive Dependency : CourseID → CourseName, faculty name

b) Normal forms
- 1NF
- 2NF (Partial dependency Exists)
- 3NF (transitive dependency Exists)

d) final decomposed Relations

1) Student (rollNo, StudentName)
2) Course (CourseID, CourseName, facultyName)
3) Enroll (RollNo, CourseID)

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Answer 8

SQL Queries

a) Department Wise Average Salary

Select Dep, AVG (Salary) from Emp group by Dept;

b) Departments With Average salary >600

select Dep from Emp Groupby Dept Having AVG(Salary) >6000;

c) Count Employee in Each Department

select Dept, count (*) from Emp group by Dept;

d) Sort Employees by Salary (Descending)

Select * from Emp ordered by salary desc;

Answer 9

Account No Int primary key,
Customer ID · INT Not NULL,
Balance Decimal (10,2) check (Balance >= 0)
Account type VARCHAR (20) Not Null.
Unique (Account No)
foreign key (customer ID) Reference customer (customer ID)
);

Answer 10

Canonical cover

a) Canonical (minimal) cover

$A \to B$
$B \to C$
$C \to D$
$D \to E$

b) Candidate key

c) Importance

Canonical cover remove redundant simplifies functional dependency and helps achieve efficient and correct normalization.